

Mini Project 1.3: Validating and Extending the 0G to 6G Comparison

1. Summary

I cross-checked headline peak data rates for five generations (2G including GPRS/EDGE, 3G, 4G, 5G, 6G) against at least two sources each. The recurring finding is that a table's single "peak rate" per generation mixes three different things: standardized theoretical peaks, minimum requirements set by ITU, and what users actually experience. I then built a keyword-based recommender that maps a plain-language requirement to the generation that first addressed it, and tested it on eight statements.

2. Annotated fact-check

Values below are the commonly tabulated headline figures. Please compare them against the exact numbers in your Section 1.2 table and adjust the "Table claim" column if yours differs.

Generation	Table claim	What sources say	Discrepancy and reason
2G (GSM, GPRS, EDGE)	about 9.6 kbps (GSM); tens to hundreds of kbps with GPRS/EDGE	GSM 9.6 kbps typical; GPRS 171.2 kbps theoretical but about 58 kbps in practice [S1, S2]; EDGE up to 384 kbps, 120–384 kbps in the field [S1]	Theoretical peaks assume all time slots and the best coding scheme for one user. A phone rarely gets them. One source lists GPRS up to 56 kbps and EDGE up to 256 kbps [S3], which reflects handset slot limits.
3G (UMTS, HSPA)	about 384 kbps to 2 Mbps	UMTS 384 kbps to 2 Mbps; the 2 Mbps is shared by all active users in a 5 MHz cell and falls to 384 kbps with just two users [S2]. HSPA theoretical 42 Mbps but averages 1–3 Mbps [S1]; another source lists HSPA up to 7.2 Mbps and 42 Mbps with dual carrier [S3]	Peak rates depend on sharing and on which release is deployed, so "3G" spans a factor of 100 or more. HSPA is itself often called 3.5G.
4G (LTE)	about 100 Mbps (LTE)	100 Mbps theoretical, but 3–10 Mbps average and 20+ Mbps peak in real use [S1]; early LTE listed at up to 75 Mbps [S3]	The gap is theory versus field measurement, and it depends on the device category and bandwidth. S1 dates from 2014, so newer LTE-Advanced networks do better. Verify against a current source.
5G (NR)	about 10–20 Gbps peak	ITU minimum requirement: 20 Gbps downlink and 10 Gbps uplink peak, 100 Mbps user-experienced downlink, 1 ms latency for URLLC, 10^6 devices per km ² [S4, S5]. M.2083 says peak is expected to reach 10 Gbps, and up to 20 Gbps under certain conditions [S6]	These are ITU evaluation requirements for candidate technologies, not measured deployment performance. The user-experienced figure (5th percentile, dense urban) is the realistic number and is 200 times smaller than the peak. Add a field-measurement source (for example a national regulator or speed-test report) to show real 5G rates.
6G	about 1 Tbps (vision)	ITU-R IMT-2030 targets are reported as 50, 100 and 200 Gbps depending on scenario, while 1 Tbps is the most common target in the research literature; suggested user-experienced rates are 300–500 Mbps, and latency 0.1–1 ms [S7]	6G is not standardized or deployed. The 1 Tbps figure is a research aspiration and ITU targets are lower, so a table should label it as a target.

3. Problems that motivated each transition

- **1G to 2G:** (a) fragmentation: incompatible national analog systems prevented roaming; (b) security: analog calls could be eavesdropped and phones cloned. GSM introduced digital voice, encryption and international roaming.
- **2G to GPRS (2.5G):** (a) circuit-switched data was inefficient, since HSCSD gave a user four or five scarce time slots and reached only 57.6 kbps [S8]; (b) no always-on connection, because users had to dial in. GPRS shared the slots among users as packet data.
- **GPRS/EDGE to 3G:** (a) data rates of tens of kbps were too low for web browsing and multimedia; (b) demand for a common global standard for mobile internet and video calls.
- **3G to 4G:** (a) real throughput was still only a few Mbps (HSPA averaged 1–3 Mbps [S1]); (b) data traffic growth needed an all-IP network with higher spectral efficiency, which LTE provided.
- **4G to 5G:** (a) 4G was not designed for 1 ms latency and high reliability, needed for control tasks such as robotics; (b) it could not support 10^6 low-power devices per km² for IoT; (c) area capacity had to rise to 10 Mbps per m² [S4]. This gave the three 5G classes eMBB, URLLC and mMTC.
- **5G to 6G:** (a) targets of 0.1 ms latency and 300–500 Mbps user-experienced rates [S7]; (b) new needs such as integrated sensing and communication and AI-native networks, plus area capacity of 30–50 Mbps per m² [S7].

4. Deliverable: rule-based recommender

The tool is a Python dictionary mapping keyword phrases to a generation and a justification. It lowercases the input, checks every keyword, and reports the matching generation with the words that triggered it. If keywords from more than one generation match, it says the requirement does not map to a single generation and lists each; if nothing matches, it asks for a more specific description. No machine learning is used.

Test input	Output	Correct?
Low-latency control for a robotic arm	5G (URLLC)	Yes
Text messages and secure, encrypted calls while roaming	2G	Yes
Browse the web and make video calls	3G	Yes
Monitor thousands of low power soil sensors	5G (mMTC)	Yes
Holographic telepresence with integrated radio sensing	6G (vision)	Yes, but 6G is not yet deployed
Analog cellular voice calls from a moving car	1G	Yes
Stream HD video and also perform remote surgery	4G and 5G (URLLC): flagged as mixed	Yes, ambiguous case handled
A better mobile experience	No match, asks for detail	Yes

Table 2: Recommender test cases and outputs.

The tool's main limitation is that plain substring matching cannot understand context: a sentence phrased with words outside the dictionary fails, and the word "sensing" would send a robotics sensor request to 6G. It is a demonstration of mapping problems to generations, not a classifier.

5. Conclusion: where real data nuances the table

The clearest case is 3G. A table saying "384 kbps to 2 Mbps" is accurate as a specification, but the 2 Mbps is shared by all active users in a cell and fell to 384 kbps with only two users [S2], and HSPA's 42 Mbps peak coexisted with 1–3 Mbps averages [S1]. The same peak-versus-typical pattern holds for GPRS, LTE and 5G, where the ITU peak is about 200 times the 100 Mbps user-experienced target [S4]. The 6G row is the most speculative, since its numbers are targets. A useful table should therefore show peak, typical and status (standardized, deployed or vision) for each generation, and the recommender should be read as pointing to the generation whose design goals first covered a problem, not one that delivers it at headline speeds.

Sources consulted

1. [S1] The Broadband Guide (SpeedGuide), "What are the actual speeds of GPRS, EDGE, UMTS, HSPA": speedguide.net/faq/what-are-the-actual-speeds-of-gprs-edge-umts-hspa-366 (2014, US-focused).
2. [S2] ITWissen, "UMTS transmission rate": itwissen.info/en/UMTS-transmission-rate-111744.html
3. [S3] Movistar community, data connection FAQ: comunidad.movistar.es (operator forum, low authority; replace if possible).
4. [S4] Wikipedia, "IMT-2020" (summary of ITU-R minimum requirements): en.wikipedia.org/wiki/IMT-2020 (tertiary; confirm against S5).
5. [S5] Nokia slides, "Requirements for IMT-2020", ITU-R WP5D: [itu.int/en/ITU-R/study-groups/rsg5/rwp5d/imt-2020/Documents/S01-1_Requirements for IMT-2020_Rev.pdf](http://itu.int/en/ITU-R/study-groups/rsg5/rwp5d/imt-2020/Documents/S01-1_Requirements%20for%20IMT-2020_Rev.pdf)
6. [S6] ITU-R Recommendation M.2083 (IMT Vision 2015), Fig. 3 and text, as reproduced at 5gitaly.eu/2018/wp-content/uploads/2018/12/5GITALY_MICHELE_ZORZI_CNIT_TALK.pdf
7. [S7] "6G: The Intelligent Network of Everything", arXiv:2407.09398: arxiv.org/pdf/2407.09398
8. [S8] ChannelPartner, "So funktioniert das mobile Internet" (HSCSD and GPRS): channelpartner.de/article/3893302/so-funktioniert-das-mobile-internet.html